

Stony Brook University
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Fall 2016

ASSIGNMENT 3

Due: Thursday, October 20 in class.

INSTRUCTIONS: Show all your work to get full points.

1. In this problem, we will explore the graphical method we saw in class to calculate the income and substitution effects. In this problem, you will work with actual utility functions and you will calculate, step by step, all the elements that are required to get a numerical value for the income and substitution effects.

(a) Consider the utility function $u(x, y) = x^{0.5}y^{0.5}$.

- i. Calculate the Marshallian demand functions, the Indirect Utility Function and the Expenditure Function.
 - ii. (Original Optimal Bundles) For the remainder of the problem, we will keep income fixed and the price of good y at $I = 100$, $p_y = 2$. Suppose that the price of good x is $p_x = 1$. Use your answers in item (i.) to calculate the optimal demands and the utility level attained at these prices and income, denote the optimal bundle by (x^A, y^A) and the utility level by U_A . Graph your answers in a diagram. Your diagram should be correctly labeled and should include the budget line, the indifference curve corresponding to utility level U_A and the optimal bundle.
 - iii. (New bundle after a price change) Now, suppose that the price of good x increases to $p_x = 1.10$. Using your results from item (i.), calculate the optimal demands and the utility level attained after this price change. Denote the optimal bundle by (x^C, y^C) and the utility level attained by U_C . Graph your answers in the same diagram as in part (ii.). Your diagram should be correctly labeled and should include the new budget line, the indifference curve corresponding to utility level U_C and the optimal bundle.
 - iv. If at the new price of $p_x = 1.10$, we would like to keep this consumer at the original utility level U_A , what is the minimum income necessary to achieve this? Use the expenditure function obtained in item (i.) to answer this question and denote the level of income that you obtain as I_A .
 - v. (Compensated Bundle) At prices $p_x = 1.10$, $p_y = 2$ and income I_A (that you obtained in the last item), use your results from item (i.) to obtain the optimal bundles and the level of utility attained. Denote the optimal bundle by (x^B, y^B) and the utility level as U_B . Is it the case that $U_B = U_A$? Graph the optimal bundle and in the same diagram we have been using. You should also graph the budget line corresponding to income I_A and prices $p_x = 1.10$ and $p_y = 2$.
 - vi. Now we are ready to calculate the income and substitution effects. The substitution effect is $SE = x^B - x^A$ and the income effect is $IE = x^C - x^B$. The total effect is, of course, $TE = SE + IE = x^C - x^A$. What are the signs of each of the effects? Does the law of demand hold, i.e. quantity demanded decreases after a price increase?
- (b) Repeat items i–vi for the utility function $u(x, y) = \sqrt{x} + \sqrt{y}$. Graph all your answers in a *new* diagram. How is the size of the substitution, income and total effect from this utility function compared to the one in part (a)?

2. Camilo gets \$3 per week as an allowance to spend any way he pleases. Because he likes only peanut butter and jelly sandwiches, he spends the entire amount on peanut butter (at \$0.05 per ounce) and jelly (at \$0.10 per ounce). Bread is provided free of charge by his concerned friend Alejandro. Camilo is a particular eater and makes his sandwiches with exactly 1 ounce of jelly and 2 ounces of peanut butter. He is set in his ways and will never change these proportions.
- (a) How much peanut butter and jelly will Camilo buy with his \$3 allowance in a week?
 - (b) Suppose the price of jelly were to increase to \$0.15 an ounce. How much of each commodity would be bought?
 - (c) By how much should Camilo's allowance be increased to compensate for the increase in the price of jelly in part (b)?
 - (d) Graph your results in parts (a) to (c).
 - (e) Discuss the results of this problem in terms of the income and substitution effects involved in the demand for jelly. *Hint: Notice that we worked in this problem with a fixed proportions (perfect complements) utility function. A consumer with these preferences is not willing to substitute goods at any other proportion. Interpret this fact in light of the income and substitution effects.*
3. Marcos drinks only pure spring water, but he can purchase it in two different-sized containers: 0.75 liter and 2 liter. Because the water itself is identical, he regards these two "goods" as perfect substitutes.
- (a) Assuming Marcos' utility depends only on the quantity of water consumed and that the containers themselves yield no utility, express this utility function in terms of quantities of 0.75-liter containers (x) and 2-liter containers (y).
 - (b) State Marcos' demand function for x in terms of p_x , p_y and I .
 - (c) Graph the demand curve for x , holding I and p_y constant at 100 and 2 respectively.
 - (d) How do changes in I and p_y shift the demand curve for x ?
 - (e) What would the compensated demand curve for x look like in this situation?